



Effect of a cyclic heating process on the CO₂/N₂ separation performance and structure of a ceramic nanoporous membrane supporting the ionic liquid 1-methyl-3-octylimidazolium tricyanomethanide

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ARTICLE INFO

Keywords:

Ceramic membranes
Ionic liquids
Heating cycles
Thermal behaviour
CO₂/N₂ separation

ABSTRACT

This work presents an investigation of the CO₂ and N₂ mixed-gas permeation rates through a supported ionic liquid membrane (SILM) developed on a ceramic composite nanofiltration substrate with a separation layer pore size of 10 nm. The ionic liquid 1-methyl-3-octylimidazolium tricyanomethanide was used as supported liquid phase. The hybrid membrane was subjected to a cyclic heating process including isothermal steps of 180 °C, under inert gas flow. In this context, the effect of temperature (over the range of 25–180 °C) and feed composition on the permeation properties and CO₂/N₂ separation performance was studied prior to and after completion of each heating cycle. For all heating cycles, the highest CO₂/N₂ selectivity values (up to 50.4) were provided at ambient temperature unlike to CO₂ permeance that was favored by elevation of temperature, reaching to $2.48 \cdot 10^{-9} \text{ mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1} \cdot \text{Pa}^{-1}$ at 180 °C. Thermal analysis by means of DSC and TGA investigations allowed to determine the physical state and thermal stability of the immobilized IL phase, thereby providing a better understanding of the determined gas permeation properties. Confinement into the nanopores of the developed SILM substantially affected the thermal behaviour and assisted decomposition of the ionic liquid phase upon heat treatment. UV/Vis spectroscopy indicated formation of functionalized by-products during prolonged periods of heating at 180 °C that could form stable chemical complexes with CO₂ molecules, thus drastically affecting both CO₂ solubility and diffusivity, and CO₂/N₂ separation efficiency.

1. Introduction

CO₂ is one of the most abundant greenhouse gases (GHGs) in the atmosphere and is largely regarded as a major cause of the observed global warming and climate change. The development of methods and materials for reducing the emissions of CO₂ has been focusing extensive research activity all over the world. The Post Combustion Capture (PCC) processes, which are based on liquid solvents that chemically absorb CO₂, are the most widely applied processes in industry for removal of CO₂ from emitted flue gas. In PCC processes, CO₂ absorption capacity, selectivity, dissolution rate, chemical stability, regeneration temperature, volatility and cost are the most important criteria for selecting liquid absorbents for CO₂ capture. Current requirements of the PCC technology are: (a) reduction of the parasitic energy load, (b) to effectively address corrosion, (c) faster absorption/stripping rates, (d)

less viscosity and less use of water, (e) to confront the problem of solvent degradation and volatility. These problems pose stimulating challenges for exploitation and development of new solvents and novel advanced materials that can pave the way for deployment of alternative efficient, sustainable, environmentally benign and cost-effective technologies for capture of CO₂ from flue gas.

A promising alternative CO₂ capture technology is based on the use of ionic liquids (ILs) as absorbents. ILs (frequently referred to as green solvents) are salts that by definition, have low melting points, i.e. below 100 °C; these salts are composed of organic cations typically having bulky and asymmetric structure, paired with organic or inorganic anions. When these salts are liquid at or below ambient temperatures, they are categorized as room-temperature ionic liquids (RTILs). ILs present some inherent advantages over the alkanolamines used in traditional flue gas scrubbers; these include extremely low vapour

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pressures (negligible volatility), broad liquid range, high thermal and chemical stability, non-flammability, corrosion inhibition properties and easy recycling [1,2]. In addition, after the first reports of an unexpectedly large CO₂ absorption capacity and selectivity in several ILs (~0.3 mol/mol at 2 MPa and 25 °C) [3], a growing interest has been focused on exploring and understanding the specific interactions between ILs and various gases [4]. The aforementioned properties make ILs attractive solvents for CO₂ capture from post-combustion flue gas, fulfilling in parallel important energy, technical and environment related aspects.

Properties such as molar volume, viscosity, polarity, and interaction strength can potentially be tailored to enhance preferential absorption and diffusion of absorbed CO₂. It has been found that CO₂ solubility can be improved by increasing the length of alkyl chains [5,6] or employing fluorinated cationic and/or anionic species [7]. The main reason is that the increase of the side chain length and the fluorine-substitution in the cations increase the IL free volume, and thus, the IL ability to accommodate more CO₂ molecules; moreover, the interaction between CO₂ molecules and the fluorinated alkyl chains is stronger, which favours physical dissolution of CO₂ [8], even though fluorination of the cation often leads to RTILs with relatively high viscosity [7]. Moreover, addition of polar functional groups such as ether linkages can improve the selectivity for CO₂/N₂ separations [9]. Carlisle et al. [10] found that the polar, terminal nitrile group of [NC-C_nmim][Tf₂N] RTILs contributed to enhanced CO₂/CH₄ and CO₂/N₂ solubility selectivity compared to analogous, non-functionalized, n-alkyl-substituted [H₃C-C_nmim][Tf₂N] RTILs. On the other hand, previous works have indicated that the nature of the counter-anion is important and can have a more pronounced effect on the CO₂ solubility and CO₂/N₂ selectivity than the nature of the cation [7,11].

In general, one of the major drawbacks of RTILs, when used as net liquid solvents, is their relatively high viscosity compared to molecular liquid absorbers. As a more convenient solution, immobilization of very thin layers of ILs on the top surface or into the pores of sufficiently permeable membranes has already evidenced the possibility to overcome the problem of slow diffusivity and improve the efficiency of CO₂ removal from flue gas [12–15]. Several techniques for immobilizing ILs on polymeric or porous inorganic supports, including physical imbibition, solvent evaporation or chemical grafting methods have been applied to develop Supported Ionic Liquid Membranes (SILMs) [16–21]. Previous testing of polymer based SILMs showed promising results with permeabilities/selectivities that were consistently above the upper bound of a Robeson plot for CO₂/N₂ separation [22]. In general, membrane-based gas separation processes offer the advantages of simple and continuous operation without need for regeneration stages, compact modular design, ease of scale-up, low environmental impact, good weight and space efficiency, lower capital costs and lower energy requirements. Moreover, compared to the conventional alkanolamine or aqueous amine-based gas separations, SILMs require much smaller quantities of the CO₂-selective solvent, i.e. IL phase to dissolve and separate CO₂ from N₂ and other gases. The performance of a SILM for CO₂ separation from N₂ (which is the major component of flue gas) is principally defined by the CO₂ permeability and separation selectivity, which are related to the solubility and diffusivity of the two gases in the supported IL phase.

If a porous inorganic SILM encompasses a non-functionalized (i.e. normal) IL, then CO₂ will dissolve in the supported IL phase by physical interaction, similar to the case of CO₂ absorption in the bulk liquid state. For inorganic SILMs with asymmetric porous structure, the actual diffusion barrier comprises the membrane layers whose pores are completely filled with the IL i.e. their total pore volume is occupied by the embedded IL phase. Therefore, these layers act as a combined dense separation layer and determine the gas permeation and separation selectivity.

In this work, a ceramic SILM has been developed via casting and imbibition of the ionic liquid 1-methyl-3-octylimidazolium

tricyanomethanide into the nanopores of an α -/ γ -alumina monolith. The employed ionic liquid exhibits high thermal stability, low viscosity and enhanced CO₂ absorption capacity compared to other imidazolium based ILs. The membrane was subjected to three consecutive heating cycles up to 180 °C with the purpose to exploit the effect of the applied heat treatment on the permeation rates of CO₂ and N₂ gases and the CO₂ separation efficiency, with respect to the untreated SILM. The separation efficiency of the SILM prior to heat treatment and after each heating cycle was evaluated for CO₂/N₂ binary mixtures with respect to the mixture composition and operation temperature. A thorough investigation was also conducted on the effect of confinement and substrate's surface chemistry (using non-porous γ -Al₂O₃) on the phase behaviour and thermal stability of the IL embedded into the nanopore structure with respect to the thermal properties and stability of the neat IL. Thermal analysis of the SILMs revealed changes in the phase behaviour of the ILs under confinement and explained the variance observed between the permeability properties of the developed SILMs.

2. Materials and methods

2.1. Chemicals and substrates

The RTIL used in this study was 1-methyl-3-octylimidazolium tricyanomethanide ([OMIM][TCM]), provided by IoLiTech GmbH. According to the specifications, the mass fraction purity of the RTIL was in excess of 98% while the water content was 120 ppm. The employed IL has exhibited relatively low viscosity and high CO₂ absorption capacity combined with superior CO₂/N₂ ideal selectivity [5,23]. It has been reported [23] that for the 1-butyl-3-methylimidazolium tricyanomethanide IL ([BMIM][TCM]), the ideal CO₂/N₂ sorption selectivity (S_H) at the temperature of 35 °C, determined from the ratio of the respective Henry's constants ($\overline{H}_{CO_2}/\overline{H}_{N_2}$), was 154, that is much higher than the S_H performances (< 100) observed up to date for ILs that interact physically with CO₂ [5]. According to MD simulations, the [RMIM][C(CN)₃] ILs exhibit unusually low shear viscosity (for the imidazolium family) due to weak cation–anion coordination and absence of hydrogen bonding between them [24].

An up-scaled ceramic multilayered ultrafiltration (UF) membrane was used as porous support to develop the SILM. The membrane (monochannel tube of 50 cm total length, 0.67 cm ID, 1 cm OD) with glazed ends (1.75 cm) was obtained from Inopor® GmbH. The UF membrane incorporated a thin mesoporous γ -alumina separation layer having a nominal pore size of 10 nm and an effective surface area of 146 cm². The separation layer, located on the outer surface of the monochannel tube, was supported by two successive macroporous α -alumina layers of intermediate pore size and a thick macroporous α -alumina substrate comprising the inner side. Mixed-gas permeance measurements were conducted using He (99.999%), N₂ (99.999%) and CO₂ (99.998%) gases.

2.2. Membrane fabrication

The supported ionic liquid membrane was developed by applying a vacuum-assisted liquid infiltration method. The membrane was accommodated into a specially designed tightly capped stainless steel module and initially degassed at 150 °C under vacuum. After lowering the temperature down to 30 °C, a certain bulk IL volume was suctioned through a throttle valve into the outer top layer side so as to completely fill the annulus space of the membrane tube within the module. The IL was allowed to wet the separation layer and intrude into the pores under the driving force of the vacuum for a period of 1 h. The hydrophilic nature of the porous γ -alumina and α -alumina layers ensured RTIL wettability and infiltration. Moreover, the relatively low viscosity of the selected IL facilitated the liquid imbibition within the pores of the treated membrane [25]. The developed SILM will be denoted herein after as SILM₁.

In addition, a second tubular ceramic multilayered α - γ - Al_2O_3 membrane of the same type as the one used to develop the SILM_1, having the same structural characteristics except for the total length (15 cm instead of 50 cm), was employed to develop a similar SILM using the [OMIM][TCM] IL. The applied infiltration procedure was the one followed for the large SILM_1. The second SILM was fabricated with the purpose to study the thermal properties of the as-produced hybrid membrane and will be denoted hereinafter as SILM_2.

2.3. Instrumentation

2.3.1. Characterization

Thermogravimetric and Differential Thermal analysis (TGA/DTA) was performed using a Perkin Elmer Pyris Diamond TG/DTA analyzer. The samples were heated under Argon flow at a rate of $3^\circ\text{C}/\text{min}$. Differential Scanning Calorimetry (DSC) analysis was conducted with the TA Instruments 2920 MDSC. In the first mode of DSC runs, the samples were quenched (with liquid Nitrogen) to $\sim -170^\circ\text{C}$ and then heated up to 330°C or 430°C at a heating rate of $10^\circ\text{C}/\text{min}$. In the second method (using the temperature-modulated DSC method), the samples were heated over the temperature range of 0 – 330°C under a heating rate of $5^\circ\text{C}/\text{min}$ with a temperature modulation $\pm 0.5^\circ\text{C}$ every 60 s. UV–vis spectra were recorded on a Hitachi U-3010 spectrophotometer equipped with an integrated sphere. Dilute solutions of [OMIM][TCM] and heat treated [OMIM][TCM] IL in propanol-2 (0.6 v/v %) were added to the cell and the spectra were recorded and corrected by subtraction from the data of the pure solvent (propanol-2) spectrum.

2.3.2. Experimental set-up for evaluation of CO_2/N_2 separation performance

Mixed-gas permeability measurements using CO_2/N_2 mixtures of varying composition as a feed were performed in the Wicke–Kallenbach (WK) configuration at the temperatures of 25, 50 and 70°C , for the as-prepared SILM. The Wicke–Kallenbach method is commonly used for studying isobaric counter-current diffusion of gaseous components through membranes and its description is given in the [Supplementary content](#). The experimental apparatus is shown in [Fig. S1 in the Supplementary content](#). The feed was introduced into the outer (i.e. separation layer) side of the SILM while He was sweeping the inner side (permeate side). The supplied CO_2 and N_2 gases were mixed before entering the membrane module and the CO_2 mole fractions were set at 0.10, 0.15, 0.2 and 0.5. The feed mole fractions of 0.10 and 0.15 lie within the range of CO_2 concentrations in real flue gas emitted from thermal power plants. The composition of the permeate and retentate stream were analyzed using a HP 5890 Series II gas chromatograph equipped with a stream selection valve, a gas sampling valve and TCD, FID detectors in series. Chromatographic separation was achieved with an Alltech Hayesep D 100/120 packed column (1/8", 30 ft). The gas flow rates were controlled by electronic mass flow controllers (Bronkhorst F-200CV) with a flow capacity of 2 – $100\text{ cm}^3\text{min}^{-1}$ (STP) for N_2 (99.999%) and CO_2 (99.998%) and 2 – $100\text{ cm}^3\text{min}^{-1}$ (STP) for He (99.999%). The temperature was controlled via a PID controller and a Type K thermocouple in contact with the outer shell of the membrane module. The applied He flow rates were $20\text{ cm}^3\text{min}^{-1}$ for all studied SILMs. The outlet flow rates were measured using a soap film flowmeter. The equations used to derive the permeance and selectivity values are provided in the [Supplementary content](#).

2.4. Thermal treatment of the developed SILM

The membrane was accommodated in a tightly capped stainless steel module, which is described in [Fig. S2 in the Supplementary content](#). The module was placed in the experimental set up used for evaluation of the gas permeance and CO_2/N_2 separation selectivity (described in Section 2.3.2). The membrane was purged with pure N_2 with

a flow rate of $20\text{ cm}^3\text{min}^{-1}$, and then subjected to three consecutive heating cycles, with the purpose to investigate the effect of the sequential thermal treatment on the membrane's permeation properties and separation efficiency. All heating cycles included an isothermal heating stage at 180°C for a specific time period; the membrane was heated up to 180°C at a slow rate of $10^\circ\text{C}/\text{h}$ under N_2 flow ($20\text{ cm}^3\text{min}^{-1}$) and then the temperature was kept constant for a period of 1 h at the 1st and 2nd cycle, and a much longer period of 24 h at the 3rd cycle. Upon completion of the isothermal step of each cycle, the membrane was slowly cooled down to 25°C at a rate of $10^\circ\text{C}/\text{h}$, and subsequently, a series of gas permeance and separation selectivity measurements was performed using the lowest CO_2 feed concentration of 6 v/v %, at the temperatures of 25, 50, 70, 100 and 180°C . In brief, the three heating cycles were carried out by repeating the same procedure (except for the duration of the isothermal step), and the three respective series of permeation measurements were conducted using the same feed mole fractions and operating temperatures. After completion of the 3rd heating cycle, mixed-gas permeability and selectivity measurements using CO_2/N_2 mixtures at the temperatures of 25, 50 and 70°C were performed for the heat treated SILM. The CO_2 mole fractions in the feed were set at 0.06, 0.15, 0.25 and 0.5. The same experimental procedure with the as-prepared SILM was applied and He was used as sweep gas.

2.5. Thermal treatment of IL-supporting non-porous γ - Al_2O_3 substrate

With the purpose to investigate the thermal behaviour and stability of the embedded IL phase, a γ - Al_2O_3 /[OMIM][TCM] sample (SILP) was prepared. Specifically, non-porous γ -alumina powder (Aluminiumoxid C, Degussa AG), having specific surface area of $100\text{ m}^2\text{g}^{-1}$, was degassed at 120°C for 24 h under vacuum (10^{-2} mbar) and impregnated with a solution of 10 v/v % [OMIM][TCM] in methanol. The quantities of the powder and the IL solution were chosen such that the entire surface of the non-porous γ -alumina particles to be covered by an ultrathin layer of IL with a thickness of 10 nm, equal to the nominal diameter of the membrane's γ -alumina separation layer pores. The mixture was magnetically stirred at 40°C for a few hours and thereafter, the stirring ceased and the remaining solvent was left to evaporate. The white γ - Al_2O_3 /[OMIM][TCM] non-porous powder was obtained and heated under vacuum at 60°C for 24 h. It is mentioned that some care should be taken at the stirring/heating and evaporation steps because the system tends to form yellowish rubber-like inverse SILPs that exhibit different thermal properties. A certain amount of the obtained SILP powder was divided into six samples; all samples but one (as-produced sample) were placed inside a tubular furnace and subjected to the same thermal treatment as the SILM. At selected time intervals the heating process was interrupted and the samples were cooled down to ambient temperature; each time one of the samples was then removed from the furnace, thereby preparing five SILP samples heated at 100°C , 140°C , 180°C (1 h), 180°C (2 h) and 180°C (24 h). The non-porous SILPs were studied by applying Thermogravimetric and Differential Thermal Analysis.

3. Results and discussion

3.1. Separation of CO_2/N_2 mixtures

3.1.1. Effect of feed composition on CO_2 permeance and CO_2/N_2 selectivity

The permeation properties of the as-prepared SILM_1 and the same membrane after being subjected to 3 heating cycles were studied at the temperatures of 25 and 70°C . As shown in [Fig. 1](#), for both untreated and heat treated SILMs, low CO_2 concentrations in the feed (i.e. ≤ 10 v/v %) resulted in a slight enhancement of CO_2 permeance with respect to the permeances determined when larger CO_2 concentrations were supplied in the feed side. For the heat treated SILM, the permeances of CO_2 at the larger gas concentrations were almost constant, indicating

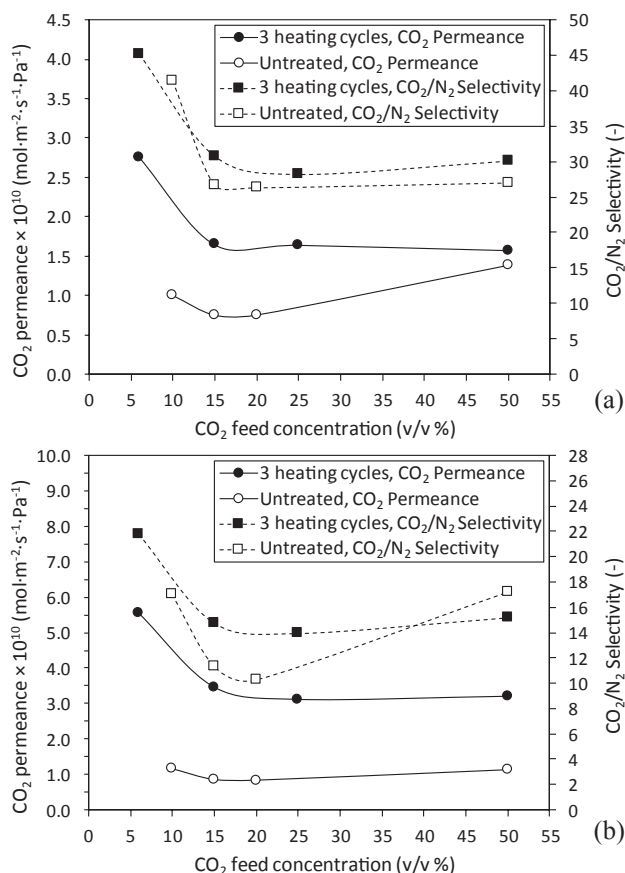


Fig. 1. CO₂ mixed-gas permeance and CO₂/N₂ selectivity vs. CO₂ feed concentration of the heat treated membrane SILM_1 (after the 3rd cycle) in comparison with the untreated SILM_1, at the temperatures of (a) 25 °C, (b) 70 °C.

that the gas permeated the membrane by the simple solution-diffusion mechanism rather than by facilitated transport. It is mentioned that the facilitated transport is distinguished by the fact that the permeance will generally decrease with increasing CO₂ partial pressure in the feed, due to saturation of the carriers (or the anionic sorption sites, as in the case of SILMs).

The independence of CO₂ permeance on feed mole fraction (> 10 v/v %) suggests that in the studied SILMs either CO₂ solubility, S , and diffusivity, D , alter towards opposite counterbalancing directions with increasing CO₂ feed partial pressure or both factors remain constant. In both cases, CO₂ permeability, P_e , which is defined as $P_e = D \times S$, is experimentally found to change tangibly with increasing CO₂ feed partial pressure. Regarding the behaviour of CO₂ solubility, confinement of the IL phase into tortuous nanopores of 10 nm width is capable of drastically affecting the absorption capacity for CO₂ due to steric effects and changes in energy of inter-ionic interaction. More specifically, nanoconfinement can induce substantial structural alterations in both individual ion pairs and mesoscopic scale, resulting in enhancement of H-bondings, Coulombic and other kinds of ionic interaction, closer ion packing, suppression of free volume and structured long-range ionic arrangements that impose reduction in gas absorption capacity when absorption proceeds by the free volume filling mechanism, compared to the bulk liquid state. As a consequence, gas solubility is likely to become insensitive to the changes of CO₂ partial pressure over a limited range of partial pressures.

In what concerns the alteration of viscosity, it has been reported that when CO₂ was dissolved in bulk 1-butyl-3-methylimidazolium tetracyanoborate ([BMIM][B(CN)₄]) at pressures up to 0.87 MPa, distinctly decreasing dynamic viscosities with increasing CO₂ mass fractions were determined, indicating weakening of the interaction among

the IL ionic species by the dissolved gas [26]. However, in opposition to the case of CO₂ absorption in the bulk liquid state, such enhancement in CO₂ diffusivity arising from solvent dilution upon gas dissolution [27], under the applied moderate partial pressures, is rather unlikely to occur considering nanoconfinement where besides steric effects, interaction with pore walls and neighboring ion pairs is stronger. In the latter case, CO₂ diffusivity would remain independent of its partial pressure in the feed in opposition to the diffusion in the bulk phase. A number of investigations [28–32] have demonstrated that confinement of ILs within capillaries of a few nanometers width favors the formation of extended H-bond networks that enhance cohesive energy and ion association. As a consequence, ILs in confined environments are forced to order in layered structures that can be markedly different from the ones of the unconfined crystalline or supercooled liquid states, and are further characterized by heterogeneous ion pair dynamics. Accordingly, MD simulation results [33–37] have manifested the decisive role of nanopore size on the resulting microstructural features (including ion packing, H-bond networks, range of spatial correlations of ionic moieties etc) and phase transitions of constrained ILs.

By comparing the CO₂ mixed-gas permeance of the membrane subjected to three heating cycles with that of the as-prepared SILM (Fig. 1), it is evident that the applied heat treatment had a favorable effect for CO₂ permeation at both ambient temperature and 70 °C, rendering the SILM more permeable for CO₂. On the other hand, CO₂/N₂ separation selectivity at 25 °C was comparable between the two studied membranes whereas at 70 °C better separation efficiency was attained by the heat treated membrane for the most of the supplied feed mole fractions. However, at both operation temperatures the two membranes exhibited the same trend in selectivity with varying feed composition; an enhancement in separation efficiency was observed with decreasing CO₂ feed mole fraction below the threshold of 15–20 v/v %. This enhancement was in agreement with the augmentation of CO₂ mixed-gas permeance at feed concentrations lower than 10 v/v %.

3.1.2. Dependence of CO₂ permeance and CO₂/N₂ selectivity on operating temperature

Increasing temperature attenuates the solubility and enhances the diffusivity of CO₂ in ILs [26]. The diffusivity increase can be attributed to the faster thermal motion of molecules and lower viscosity of the IL. Regarding the membrane subjected to 3 heating cycles, CO₂ permeance was enhanced when the temperature increased from 25 to 70 °C, regardless of the CO₂ concentration in the feed stream, as shown in Fig. 2. This trend indicates that the increase in CO₂ diffusivity in the embedded IL phase is larger than the reduction in solubility, thereby resulting in the observed enhancement of CO₂ permeance. It has been reported that the absorption of CO₂ in neat 1-butyl-3-methylimidazolium tricyanomethanide ([BMIM][TCM]) IL showed an activation energy for

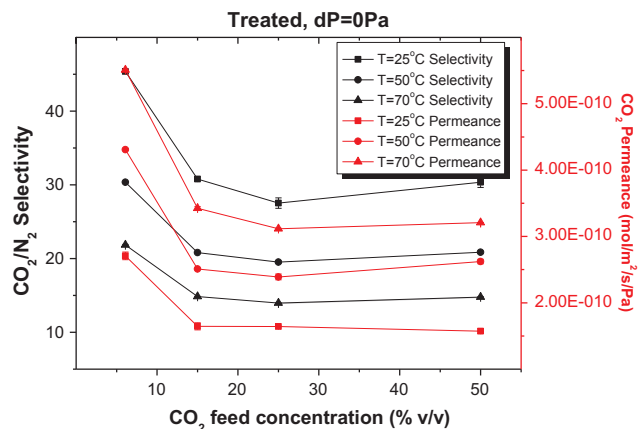


Fig. 2. CO₂/N₂ selectivity and CO₂ mixed-gas permeance of the heat treated SILM_1 (3rd cycle) with respect to the CO₂ feed concentration at the temperatures of 25, 50 and 70 °C.

diffusion as high as $41 \text{ kJ} \cdot \text{mol}^{-1}$ over the range of $18\text{--}50^\circ\text{C}$ [23]. Moreover, the viscosity of pure [BMIM][TCM] dropped by 45% in the temperature range of $20\text{--}50^\circ\text{C}$. The relation of CO_2 diffusivity normalized with temperature and solvent viscosity has been investigated for several imidazolium, ammonium, phosphonium and pyridinium ILs [38–41]. It is mentioned that the data for diffusion of gases in ILs reported so far indicate a fractional Stokes-Einstein behaviour having the form given by: $D \propto (T/\eta)^m$, where for the exponent m it is generally $0.3 < m < 0.7$. However, these studies have pointed out that the power dependency of diffusivity is different for different classes of ionic liquids. Morgan et al. [42] used a Lag-Time technique to measure gas diffusivities in imidazolium based ILs and found a power of $m = 0.6$.

However, a question might arise as to whether further temperature increase would continuously retain the increasing trend of CO_2 permeance up to the onset of the thermal decomposition of the IL phase or the monotonous decline in gas solubility would surpass the respective enhancement in diffusivity as soon as a temperature threshold is reached, thereby inducing an inversion in the temperature dependency of gas permeance prior to commencement of thermal degradation.

Mixed-gas permeance and CO_2/N_2 selectivity measurements were also conducted after the membrane had been subjected to the 1st, 2nd and 3rd heating cycle, that is each individual cycle was followed by the respective series of measurements, where the CO_2 feed concentration was 6 v/v % (Fig. 3). In all cases, no pressure difference was applied across the two membrane sides ($\Delta P = 0$). As depicted in Figs. 3(a) and 4, for each temperature, the membrane provided the largest CO_2 permeances after the 1st heating cycle. Thereinafter, the SILM was rendered less permeable for CO_2 when subjected to further heat treatment.

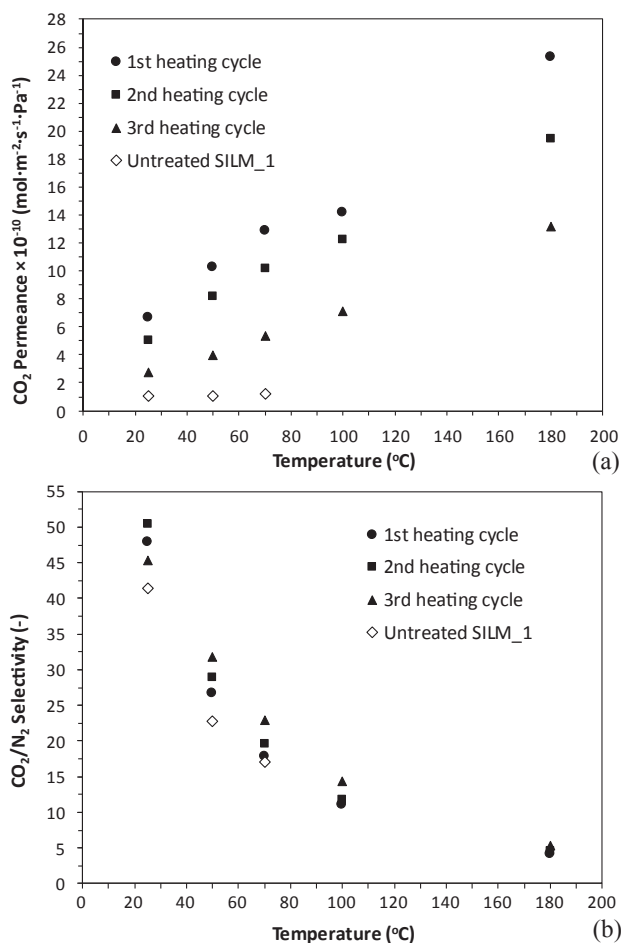


Fig. 3. Effect of the heating cycles on (a) CO_2 permeance and (b) CO_2/N_2 selectivity over the temperature range $25\text{--}180^\circ\text{C}$ for the membrane SILM_1. CO_2 feed concentration was 6% v/v.

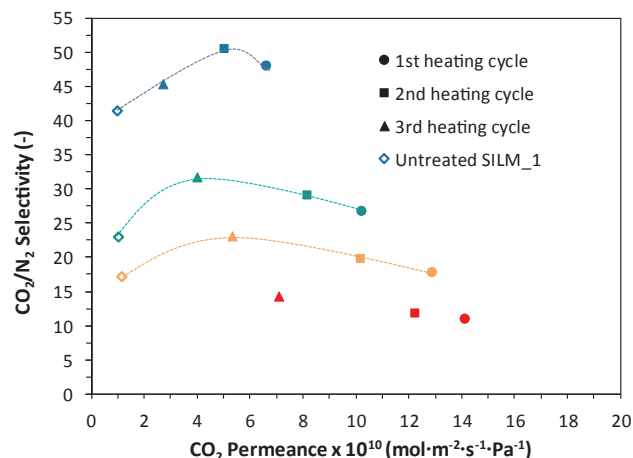


Fig. 4. CO_2/N_2 selectivity vs. CO_2 mixed-gas permeance of the heat treated SILM_1 after each heating cycle; coloured symbols were used to distinguish the temperatures of the measurements i.e. blue (25°C), green (50°C), orange (70°C) and red (100°C). CO_2 feed concentration was 6% v/v. Coloured lines have been added to guide the eye. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

On the other hand, the application of 3 heating cycles led to slightly higher CO_2/N_2 selectivities over the range of $50\text{--}180^\circ\text{C}$, with respect to the two previous cycles, as shown in Fig. 3(b). It is also noted that at 25°C the membrane was most selective (i.e. 50.7) after completion of the 2nd cycle.

Moreover, for all heating cycles, the increasing trend of CO_2 permeation rates with rising temperature continued up to the temperature of the isothermal step (180°C), thereby suggesting a clear predominance of diffusivity augmentation of CO_2 over the respective suppression of solubility over the entire temperature range, even though there are no experimental data for CO_2 permeance between 100 and 180°C . It is mentioned that after conducting the 1st, 2nd and 3rd cycle, the CO_2 mixed-gas permeance (for CO_2 feed concentration of 6 v/v %) was increased from $6.62 \cdot 10^{-10}$, $5.06 \cdot 10^{-10}$ and $2.75 \cdot 10^{-10} \text{ mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1} \cdot \text{Pa}^{-1}$ at the 25°C to $2.48 \cdot 10^{-9}$, $2.01 \cdot 10^{-9}$ and $1.35 \cdot 10^{-9} \text{ mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1} \cdot \text{Pa}^{-1}$ at 180°C respectively. It is mentioned that the CO_2 permeance for the untreated SILM at 25°C and $\Delta P = 1 \text{ bar}$ was $3.45 \cdot 10^{-10} \text{ mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1} \cdot \text{Pa}^{-1}$ when a CO_2 feed concentration of 7 v/v % was supplied [43], that is equal to 52% of the value determined after the 1st heating cycle.

On the other hand, the substantial enhancement of CO_2 permeation rates with increasing temperature was accompanied by drastic reduction of CO_2/N_2 separation efficiency (Fig. 4(b)); for example, after completion of the 1st, 2nd and 3rd cycle the CO_2/N_2 selectivity was reduced from 48, 50.7 and 45.4 respectively at 25°C down to 4.2, 4.6 and 5.3 at 180°C . This trend was common to the case of the untreated SILM where the CO_2/N_2 selectivity was also reduced when the temperature increased from 25 to 70°C , as shown in Fig. 1. The maintenance of a stable temperature dependency for CO_2 permeance and CO_2/N_2 selectivity after completion of three successive heating cycles is indicative of an unaltered gas transport mechanism through the diffusion barrier of the SILM_1. For example, assuming the limiting case where the embedded IL phases had been completely evaporated or decomposed during the thermal treatment and the separation layer pores with a nominal width of 10 nm were completely open, a predominant Knudsen diffusion for both gases would be anticipated; in this case, the opposite temperature dependency for CO_2 permeance would be observed. In fact, for open cores larger than 2 nm i.e. if $2 \text{ nm} < w < 10 \text{ nm}$, the heat treated SILM would exhibit gas transport properties corresponding to Knudsen diffusion and the ideal CO_2/N_2 selectivity would be ~ 0.8 at all temperatures, which is much lower than the experimentally determined selectivity values. Consequently,

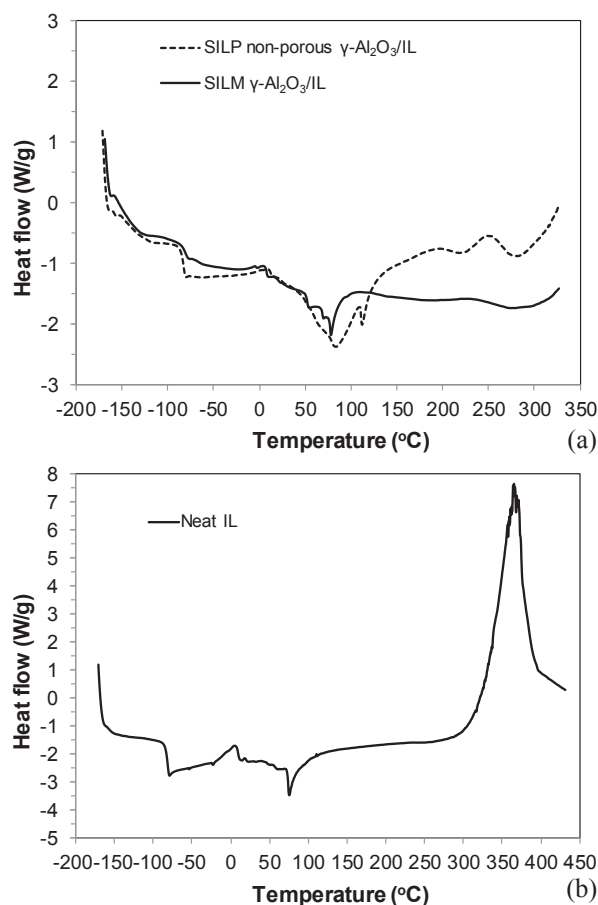


Fig. 5. DSC thermographs after quenching of the (a) non-porous SILP and SILM_2 sample, (b) neat [OMIM][TCM] IL.

elucidation of the physical state, chemical structure and thermal stability of the immobilized IL phase prior to and after completion of the heating processes is necessary in order to provide a better understanding of the experimentally determined permeation properties.

3.1.3. Phase behaviour and thermal stability of the immobilized RTIL

3.1.3.1. DSC and MDSC experiments. To investigate the thermal behaviour and stability of the confined IL phase, we have performed DSC and MDSC runs of the IL (i) in the neat liquid state, (ii) after immobilization on the surface of non-porous γ - Al_2O_3 particles, having a nominal BET surface area of $100 \text{ m}^2\text{g}^{-1}$ and (iii) after embedment into the 10 nm pores of the separation layer of the SILM_2. The later sample was obtained by carefully scratching the thin top layer on the inner side of the membrane. The second sample (SILP) was in a powder form (prepared according to the description provided in the Section 2.5) and was studied as produced, i.e. without any thermal treatment.

The samples were quenched down to -170°C and then heated up to 330°C (solid samples) or 430°C (neat IL) at a rate of 10°Cmin^{-1} . The thermograph of the neat IL [OMIM][TCM] showed only a glass transition occurring at -84.0°C ($T_{g,\text{onset}}$), and no further peaks corresponding to crystallization and melting (Fig. 5). The lack of these phase transitions in the neat IL thermograph might originate from insufficient packing of the ion pairs on the cooling step, as a result of the bulky asymmetric structure of their ionic components and absence of strong inter-ionic interaction and hydrogen bondings. Glass transition was also observed in the thermographs of the non-porous SILP and the SILM_2 sample; the transition occurred at similar temperatures to that found for the neat IL, with $T_{g,\text{onset}} = -85.8$ and -83.6°C respectively (Table 1).

On the other hand, it is mentioned that neither the SILP material nor the SILM_2 sample gave any exotherm or endotherm peaks that could

Table 1

DSC results for the neat [OMIM][TCM] IL, the SILP sample (non-porous γ - Al_2O_3 /IL) and the SILM_2 sample (γ - Al_2O_3 /IL, 10 nm pore size).

Sample	Glass transition		Exotherm peak	
	$T_{g,\text{onset}} (^\circ\text{C})$	$T_{g,\text{midpoint}} (^\circ\text{C})$	$T_{g,\text{onset}} (^\circ\text{C})$	$T_{\text{peak}} (^\circ\text{C})$
Neat IL	-84.0	-81.9	254.5	364.8
SILP	-85.8	-83.3	218.8	248.4
SILM_2	-83.6	-80.6	191.3	229.5

be attributed to crystallization and melting of the IL phase respectively (Fig. 5(a)). The endotherms from $\sim 70^\circ\text{C}$ up to 90 – 100°C can be attributed to removal of moisture. However, at higher temperatures, the non-porous SILP presented an exotherm peak with $T_{\text{onset}} = 218.8^\circ\text{C}$ and a peak at 248.4°C . Directly after the end of the exotherm peak at 279.8°C , a second exotherm peak seemed to arise up to the upper temperature of the run at 327°C . Similar behaviour was exhibited in the thermograph of the SILM_2 sample (Fig. 6(a)), which showed a broad exotherm peak within the high temperature range, with $T_{\text{onset}} = 191.3^\circ\text{C}$ and a peak at 229.5°C . This broad peak was followed by the onset of a second exotherm peak after the end of the first exotherm at 273.5°C . As will be discussed next, these exotherms cannot be attributed to phase changes in the two samples and reveal successive thermally induced structural transformation stages of the immobilized IL phase. It is mentioned that the thermograph of the neat IL presented a very intense exotherm at temperatures higher than 252°C , with a peak at 364.8°C , as shown in Fig. 5(b). This temperature was beyond the upper limit of the temperature range of the DSC runs for the two solid samples mentioned above.

The modulated DSC runs, presented in Fig. 6, revealed that both endotherms and exotherms in the thermographs of the non-porous SILP and the SILM_2 sample arise not from reversible phase changes but from non-reversible phenomena, specifically removal of water content and thermal degradation reactions.

3.1.3.2. TGA/DTG and DTA results. In order to study the thermal stability of the neat IL and the solid samples, and also verify the possibility that the DSC exotherms that emerged at higher temperatures originate from thermal structural transformation of the IL, Thermogravimetric (TGA) and Differential Thermal Analysis (DTA) were performed for all the examined samples. The weight loss curves, their dynamic derivatives (Differential Thermogravimetry, DTG) and DTA results are depicted in Fig. 7. The overall TGA results concerning the mass loss curves and DTG curves of the studied samples are presented in Fig. 8. For the solid samples, three consecutive mass loss stages (producing the respective DTG peaks) occurred at $T > 170$ – 180°C , with the second stage being the most intense. As is observed, the temperature range where the first DTG peak emerged is close to the range of the (non-reversible) exotherm DSC peak for both solid samples (see Table 1). For this peak, the relative mass loss, $\Delta m_{\text{relative}} \%$, (defined here as $(\Delta m / \Delta m_{\text{total}}) \times 100\%$, where $\Delta m = m_{\text{peak onset}} - m_{\text{peak end}}$ and $\Delta m_{\text{total}} = m_{\text{initial}} - m_{\text{remaining}}$; peak onset and peak end temperatures were identified by the DTG curves) is minor for both solid samples; however, it is indicative of decomposition owing to inter-ionic reactions taking place in the IL phase. At this point it is noticed that the mass loss fraction related to the γ -alumina support alone is negligible over the entire range of temperatures, as shown in Fig. S3 in the Supplementary content.

The maximum of the second DTG peak for the two solid materials was detected at 316°C (SILP) and 324°C (SILM) that is close to the maximum of the first peak in the DTG plot of the neat [OMIM][TCM] IL (327°C), as observed in Fig. 8(b). Taking into account this proximity, one can observe that the temperature range of the 1st and 2nd DTG peak for the two solid samples, that is approximately from 172 to 376°C , is roughly identical to the range where the intense 1st DTG peak

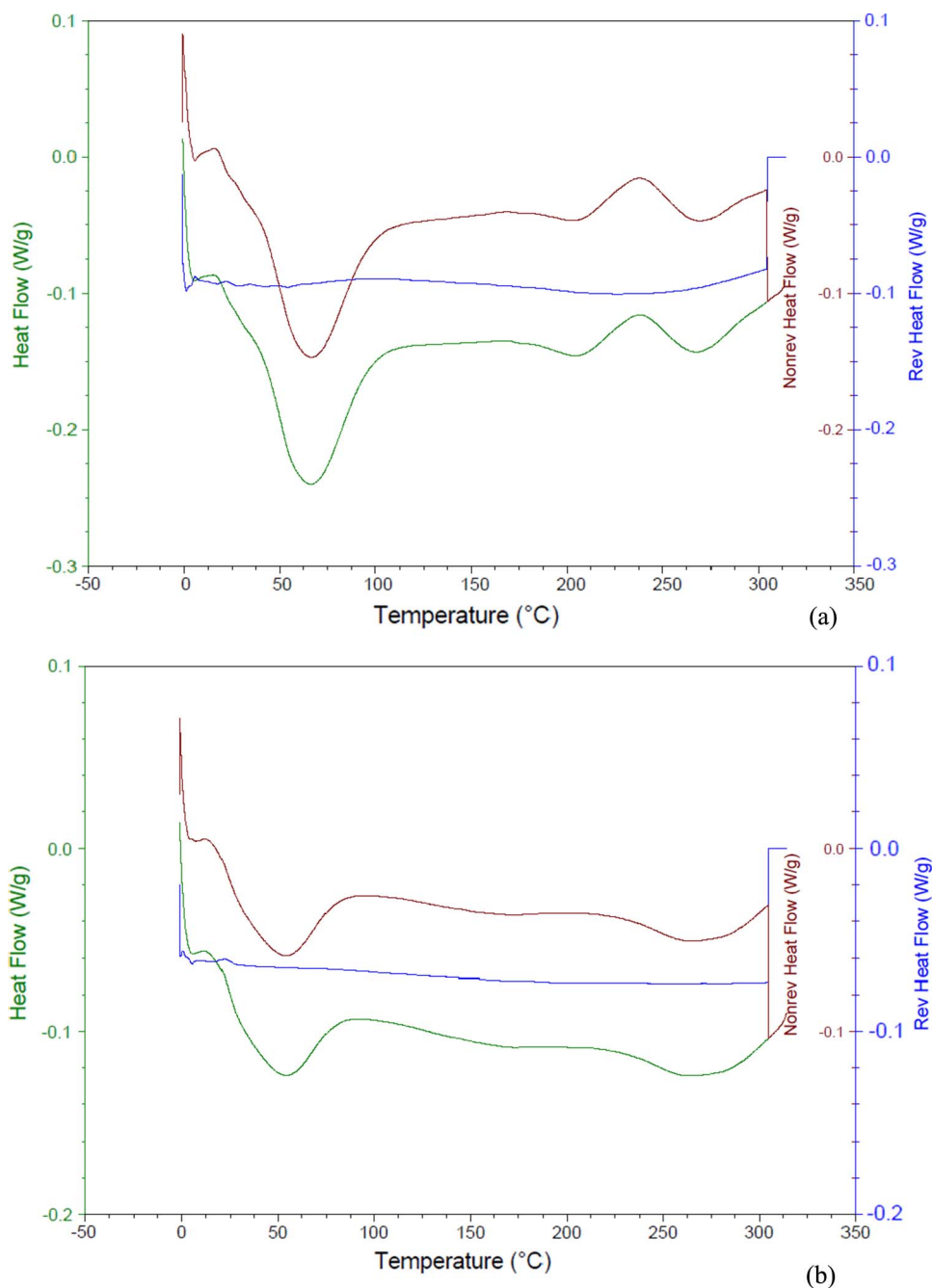


Fig. 6. MDSC thermographs of the (a) non-porous SILP material and (b) SILM_2 sample.

of the neat IL emerged (i.e. from 185.8 to 370 °C, even though for temperatures below 230 °C the IL mass loss is a quite slow process, as shown in Figs. 7(a) and 8(a). The coincidence of these temperature ranges as well as the proximity between the maximum temperatures of the 2nd DTG peak of the solid materials and the 1st peak of the neat IL, suggest different decomposition reaction pathways taking place in the IL phase, when the IL is immobilized on the surface of the non-porous γ -alumina substrate and within the porous γ -alumina membrane sample. It is likely that certain degradation reactions giving rise to the initial peak, take place over the range of 178–263 °C (SILP) or 173–250 °C (SILM) whereas other degradation mechanisms being responsible for the 2nd peak follow thereafter up to the temperatures of 375–376 °C. The earlier onset of the 1st DTG peak in the studied SILM material indicates slight facilitation for certain degradation reactions arising from the confinement of the IL phase within the nanopores of 10 nm

width that assists degradation in addition to the catalytic effect of the surface of the γ -alumina layer nanopores.

According to the results presented in Table 2, the TGA analysis of the neat IL showed a remaining weight of 21.87% of the initial sample weight, at 609.33 °C. Furthermore, it would be useful to estimate the weight ratio (%) of the supported IL phase with respect to the non-porous γ -alumina and the porous SILM substrate at the beginning and over the entire heating range up to the highest temperature limit of the TGA analysis. This estimation would allow to calculate the respective mass loss curves of the supported IL phase for each of the two studied solid samples, decoupled from the mass loss of the support. The weight loss of the support is negligible and is not taken here into account for simplicity, since (i) for the pristine γ -Al₂O₃ it was found not larger than 0.18% at 604.89 °C (see Fig. S3 in Supplementary content) and (ii) the thermal behaviour and stability of the γ -alumina substrate is presumed

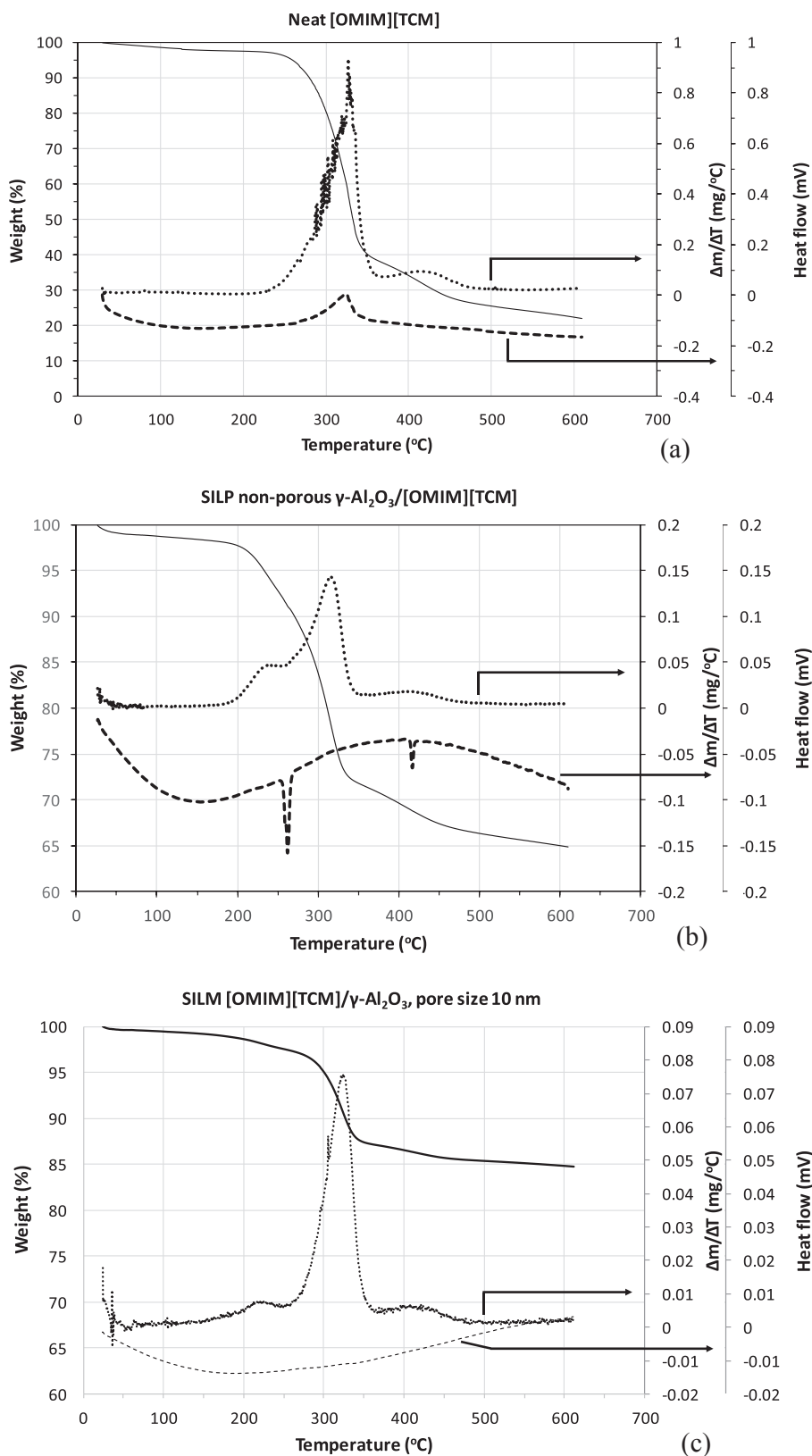


Fig. 7. TGA, DTG and DTA results of the (a) neat [OMIM][TCM] IL, (b) non-porous SILP material and (c) SILM₂ sample.

to be unaffected from the degradation of the supported IL phase (inert support).

To proceed with these calculations, it has been assumed that all structural transformation or decomposition reactions in the IL phase

have already been completed when the highest temperature limit is reached and only minor changes in the sample weight and composition might be anticipated at higher temperatures, attributed to ongoing carbonization processes. Thus, regarding the two solid samples, for the

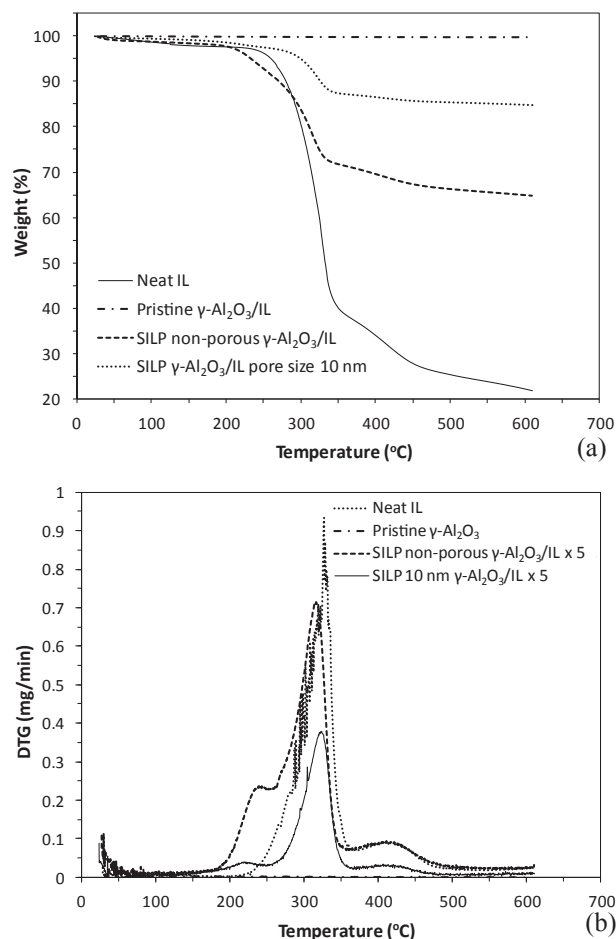


Fig. 8. Overall TGA results of the studied samples, (a) mass loss curves (b) DTG curves.

supported IL phase, a remaining weight of 21.87% of the (unknown) initial weight of the immobilized IL is presumed, in line with the case of the neat IL which showed the same final weight. The final residual mass ratio corresponds to the amount of an extensively carbonized phase [44].

In the case of the non-porous SILP material, the total remaining weight was 64.84% (see Table 2). If a and b are the unknown masses (in mg) of the IL phase and inorganic support respectively in 100 mg of the SILP material, then at the highest temperature limit it holds that $0.2187a + b = 64.84$, as well as $a = 100 - b$. Thus, in the total final mass of the material, i.e. 64.84 mg, the mass of the non-porous $\gamma\text{-Al}_2\text{O}_3$ substrate is 54.998 mg, which in these calculations is regarded as constant over the entire temperature range, and the remaining (degraded) IL phase has a mass of 9.842 mg. By subtracting the constant weight (normalized mass), w_s (%), of the support (54.998%) from the

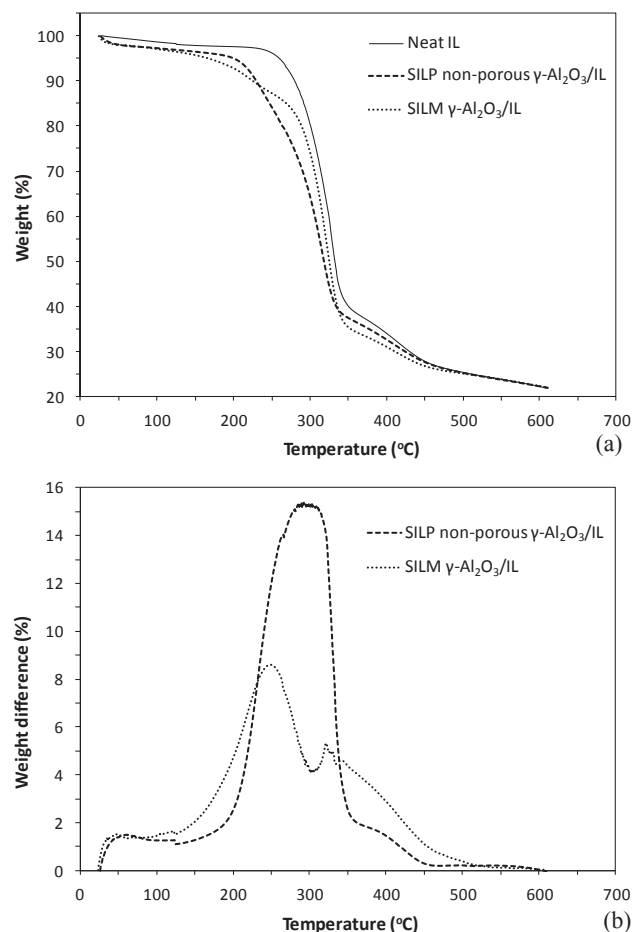


Fig. 9. Overall TGA results of the IL phase, (a) mass loss curves (b) difference between the normalized mass of the pure IL and normalized mass of the embedded IL phase.

initial weight ratio (100%) of the sample, the initial weight, $w(T_o)_{sl}$ (%), of the supported IL phase is derived, i.e. 45.002%. Therefore, if $w(T)_{sl}$ is the weight (%) of the sample at every T , then the corresponding weight (%) of the supported IL phase is calculated as $[(w(T)_{sl} - 54.998)/45.002] \times 100$ (%). This method allows to calculate the “decoupled” mass loss curves of each one of the two solid materials that present solely the weight change of the supported IL phase with rising temperature, as depicted in Fig. 9(a). Moreover, according to the aforementioned calculations, for the SILM_2 sample it is: $w_s = 80.49\%$ and $w(T_o)_{sl} = 19.51\%$.

An interesting finding is that the thermal stability of the IL was deteriorated by the presence of the two $\gamma\text{-Al}_2\text{O}_3$ supports. The strong effect imposed by the non-porous $\gamma\text{-Al}_2\text{O}_3$ substrate was probably due to its significant specific surface area ($100\text{ m}^2\text{g}^{-1}$). The difference

Table 2
TGA/DTG results for the neat IL, the SILP material and the SILM_2 sample.

Sample	Neat IL			SILP			SILM_2		
	1st peak	2nd peak		1st peak	2nd peak	3rd peak	1st peak	2nd peak	3rd peak
Relative mass loss (%)	77.0185.8327.0	16.3		18.8	55.6	12.3	9.5	69.3	9.5
$T_{o,deg}$ (°C)		370.0		178.1	262.9	375.6	172.8	250.3	375.0
T_{peak} (°C)		414.7		238.8	316.0	411.1	219.3	324.0	420.7
Total mass loss (%) ^d	78.13 ^a			35.16 ^b			15.24 ^c		

^a At $T_{max} = 609.33$ °C.

^b At 610.05 °C.

^c At 610.42 °C.

^d Total mass loss (%) = $[100 - [(m_{remaining}/m_{initial}) \times 100]]$ (%).

between the weight of the neat IL and that of the supported IL (in the SILP and SILM_2 sample) at each temperature T can provide quantitatively the excess of weight of the decomposed amount of IL caused solely by the support at any particular temperature. Thus, if the support had no catalytic effects on IL thermal degradation, the difference would not exist and the “decoupled” mass loss curves of the IL in the solid samples would be identical to the neat IL mass loss curve, presuming constant weight for the γ -alumina support over the entire temperature range of the TGA analysis. The excess of weight loss of the decomposed IL at any temperature, $w(T)_{ls}$ (%) is calculated by subtracting the weight of the neat IL, $w(T)_{nl}$ (%) from the weight of the supported IL phase at each temperature, $w(T)_{pl}$ (%); the later is determined by the ratio of the mass of supported IL at a given T to the initial mass of the supported IL phase, $m_{pl}/m_{o,pl}$. Thus, the excess of weight loss is calculated as: $w(T)_{ls}$ (%) = $w(T)_{nl}$ (%) – $(m_{pl}/m_{o,pl}) \times 100$, and it follows that:

$$w(T)_{ls}(\%) = \left(\frac{m(T)_{nl}}{m(T_o)_{nl}} \right) \cdot 100 - \left[\frac{\left(\frac{m(T)_{sl}}{m(T_o)_{sl}} \right) - \left(\frac{m_s}{m(T_o)_{sl}} \right)}{1 - \left(\frac{m_s}{m(T_o)_{sl}} \right)} \right] \cdot 100 \quad (1)$$

or

$$w(T)_{ls}(\%) = \left(\frac{m(T)_{nl}}{m(T_o)_{nl}} - \frac{m(T)_{sl} - m_s}{m(T_o)_{sl} - m_s} \right) \cdot 100 \quad (2)$$

where $m(T)_{nl}$, $m(T)_{sl}$ and $m(T)_s$: mass of the neat IL, solid/IL sample and solid substrate respectively at any temperature T ; $m(T_o)_{nl}$, $m(T_o)_{sl}$ and $m(T_o)_s$: initial mass of the neat IL, solid/IL sample and solid substrate respectively (onset of the TGA analysis). The mass of the supported IL at a given T is derived by subtracting the mass of the solid substrate from the mass of the solid/IL sample, i.e. $m(T)_{pl} = m(T)_{sl} - m_s$.

The $w(T)_{ls}$ (%) vs. T plot is presented in Fig. 9(b). As observed, for the non-porous SILP the maximum excess of weight loss is larger and found over the range of 287–315 °C, whereas for the sample derived from the SILM_2 sample, two distinct though shorter peaks appeared at 249.4 and 321.6 °C. This plot actually reflects the intensity of the catalytic effect of the γ -alumina substrate on the decomposition of the supported IL phase. However, it can also provide indication for the mesoscopic structure of the immobilized IL. In the case of the non-porous SILP, the single $w(T)_{ls}$ peak between 180 and 280 °C indicated increasing contribution from surface-catalyzed reactions taking place in parallel with purely thermal ones. It is mentioned that the rising part of the $w(T)_{ls}$ peak and the exotherm peak in the DSC thermograph share the common temperature range between 219 °C and 280 °C (see Fig. 5 and Table 1), while the intense exotherm peak related to the thermal decomposition of the neat IL commenced slowly from 254.5 °C. Therefore, the exotherm peak in the DSC thermograph of the SILP material can be attributed to exothermic surface-catalyzed decomposition reactions of the supported IL phase.

Regarding the SILM_2 sample, the doublet peak $w(T)_{ls}$ possibly originated from the fact that the immobilized IL is structured in two phases with different ion pair-pore wall interaction and degree of ordering. The first phase includes the ion pairs confined within mesopores of 10 nm size, with larger concentration and long-range ordering at the interface, which experience enhanced weak van der Waals interactions and long range electrostatic forces as well as stronger interaction with the pore walls that could facilitate decomposition reactions. The second phase is the IL adsorbed on the external surface of the separation layer and includes all the layers of loosely packed ion pairs, which tend to approach the bulk phase properties, as their distance from the external surface increases.

The formation of IL phases with different degree of ordering when embedded into nanopores has been also predicted by Pinilla et al. [45], using MD simulations for imidazolium based ILs confined between parallel structureless walls of distances. It was revealed that an interfacial layer was formed extending from the pore wall up to the first density minimum, consisting on an IL phase with excess of cations.

Density profiles perpendicular to the walls showed an interfacial liquid layer twice as dense as the bulk whereas a major part of the layering was due to the distribution of anions. The concentration of ions at the interface changed with the interwall distance, presenting a maximum at 2.8 nm for a distance between the plates of 4.5 nm.

Considering the stronger interaction with the pore walls and the higher concentration of ion pairs (or higher density of the enclosed IL layers) at the solid–liquid interface, it is plausible to assume that the first $w(T)_{ls}$ peak at lower temperatures represents the confinement-enhanced catalytic effect of the surface chemistry on the thermal degradation of the supported IL phase, whereas the second $w(T)_{ls}$ peak is due to the catalytic effect of the surface chemistry alone. Also, in Fig. 9(a) and (b) it can be noticed that at temperatures higher than 450 °C the catalytic effect of the γ -alumina substrate (expressed by the $w(T)_{ls}$) becomes equal or less than 1%; thereafter, the pyrolysis of the supported IL proceeds through purely thermal reaction mechanisms, in similarity to the thermal degradation of the neat IL.

3.1.3.3. UV/Vis results. In order to study the chemical stability and possible structural alterations of the neat IL subjected to the same heating cycles applied for the non-porous γ -Al₂O₃/IL samples (Section 2.5), UV/Vis spectra were collected for certain IL samples diluted with 2-propanol. The examined liquid samples were pure 2-propanol (solvent), an as-prepared IL sample (25 °C), two samples containing IL previously heated at 100 °C and 140 °C, and three samples containing IL previously heated at 180 °C for 1 h, 2 h and 24 h. As observed in Fig. 10, the broad absorption band at wavelengths less than 315–320 nm, became less intense and narrower as the neat IL was heated up to 180 °C for 1 h and then became again broader and stronger when the heating was prolonged to 2 h and 24 h. The derived UV–Vis spectra in the regions of 190–450 nm and 280–400 nm are presented in more detail in Fig. S4 in the Supplementary content. As observed in Fig. S3(b), the low intensity bands with maximum absorbance between 335 and 345 nm followed the same temperature dependency as the intense broad bands in the UV region of 190–320 nm. The weak bands were minimized with rising temperature and then became more intense and clear when the isothermal heating step at 180 °C lasted for 2 and 24 h. In both absorption bands, this variation can be attributed to possible changes in purity and chemical structure of the neat IL by the heating process. Burrell et al. [46], stated that for the ionic liquid 1-butyl-1-methylpyrrolidinium bis(trifluoromethanesulfonyl)amide, significant absorption above 300 nm is an indication of existence of coloured impurities. Although elucidation of the reaction mechanisms and kinds of by-products formed during heat treatment of the [OMIM][TCM] IL was not an objective of this work, it can be assumed that heating the neat IL up to 180 °C for a period of 1 h thoroughly removed organic

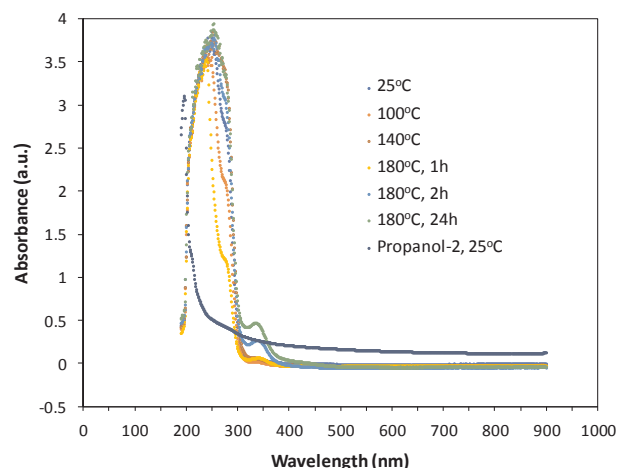


Fig. 10. UV–Vis spectra of untreated and heat treated IL solutions.

hypsochromic impurities whereas elongation of the isothermal step duration at 180 °C caused formation of coloured by-products that enhanced the absorbance of the respective samples.

It is mentioned that a non-trivial weight loss of 5.7% of the supported IL on the SILM at 180 °C was caused by thermal degradation reactions with a minor contribution from parallel surface-catalyzed reactions (Fig. 9). It is also reminded that the SILM provided the highest mixed-gas CO₂ permeances after heating at 180 °C for 1 h (i.e. 1st cycle), as observed in Fig. 3(a). On the other hand, the obtained CO₂/N₂ selectivities were most enhanced by applying 3 heating cycles to modify the membrane. This suggests that the embedded ion pair components are likely to form functional groups (such as alkyl imine groups) having high affinity for CO₂, as a result of thermal degradation reactions taking place at 180 °C [47]. Moreover, in the case that the released chromophores are conjugated, they lower the energy of electron transitions, thereby broadening the absorption band in the UV region towards higher wavelengths. If the population of these functional groups (chromophores) increases with heating for relatively extended periods, the functionalized ionic species could provide strong chemical interactions with CO₂ molecules that would drastically increase viscosity and reduce diffusivity, thereby decreasing both CO₂ and N₂ permeation rates. A number of investigations has clearly demonstrated that CO₂ solubility can be remarkably enhanced via stronger chemical interaction in functionalized ILs incorporating amine groups or other functional groups into their cation or anion [48–50]. On the other hand, the high viscosity of these ILs that strongly suppresses CO₂ diffusion is further increased by formation of a thermally stable carbamide complex. However, the parallel increase in CO₂ solubility would enhance the CO₂/N₂ selectivity especially at moderate temperatures (e.g. 50–100 °C) where chemisorption of CO₂ can predominate over physical absorption [43].

4. Conclusions

In the current work, a ceramic SILM has been developed via casting and imbibition of the ionic liquid 1-methyl-3-octylimidazolium tricyanomethanide ([OMIM][TCM]) into the porous structure of an α -alumina monolith bearing a γ -alumina mesoporous separation layer having pore size of 10 nm. The membrane was subjected to three consecutive heating cycles up to 180 °C with the purpose to exploit the effect of the applied heat treatment on the permeation properties and the CO₂ separation efficiency, with respect to the as-prepared SILM. Moreover, a SILP sample (non-porous γ -Al₂O₃/[OMIM][TCM]) and a sample derived from an identical SILM (denoted as SILM₂) were also prepared.

The heating at 180 °C for a period of 1 h provided the highest CO₂ mixed-gas permeances, which declined with increasing the total duration of heat treatment, whereas prolonged heating had a favoring effect on CO₂/N₂ selectivity. The permeation properties and the CO₂/N₂ separation efficiency of the as-prepared SILM and the same membrane after been subjected to extended heat treatment (3 cycles with totally 26 h of isothermal heating at 180 °C) were compared. For all the studied membranes, increasing temperature favored CO₂ permeation but deteriorated their separation selectivity. The two membranes exhibited the same trend in selectivity with varying feed composition; the separation efficiency was enhancement with decreasing CO₂ feed mole fraction below the threshold of 0.15–0.2.

DSC runs in the SILP material and a sample derived from the SILM₂ showed exotherms that suggested successive thermal decomposition stages of the immobilized IL phase. The thermal stability of the neat IL was relatively high compared to most of the common ILs. A detailed elaboration of the TGA analysis results revealed a confinement-enhanced catalytic effect of the surface chemistry on the thermal degradation of the supported IL phase. It was deduced that the confinement of the IL phase within the nanopores of 10 nm width provided slight facilitation for surface-catalyzed reaction pathways leading to IL

structural transformation prior to the degradation of the IL adsorbed phase on the external surface of the separation layer that included all the layers of loosely packed ion pairs.

UV–Vis spectra in diluted heat treated neat IL samples suggested the possibility that the ion pair components can form functional groups owing to thermal degradation reactions taking place at 180 °C. The functionalized ionic by-products could chemically capture CO₂ molecules, thus forming stable molecular complexes that would drastically increase viscosity and reduce diffusivity; as a result, both CO₂ and N₂ permeation rates would be reduced. However, the parallel increase in CO₂ solubility would enhance the CO₂/N₂ selectivity especially at moderate temperatures where chemisorption of CO₂ can predominate over physical absorption.

Acknowledgements

This work was supported by the Industrial Research Fellowship Program, which is granted by the Stavros Niarchos Foundation, Greece.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.seppur.2018.02.013>.

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